

Francesco Vicidomini

Software Development Specialist with 6+ years of experience designing Angular and TypeScript applications for cybersecurity, European research, and public-sector platforms. Specialized in reactive architectures with RxJS and NgRx, with hands-on experience in technical decision-making, team coordination, mentoring, and international project collaboration.



CONTACT

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CORE SKILLS

Core Frontend

Angular TypeScript
RxJS NgRx Nx
Tailwind CSS

Web Platform & Quality

REST APIs WebSocket
React / Next.js (portfolio)
Jest Unit Testing

Backend

Node.js Spring Boot
MongoDB MySQL

DevOps & Practices

Git GitLab Docker
CI/CD AWS
Agile Scrum

Domains

Cybersecurity
Privacy Risk Assessment
Digital Twin
Public Administration

LANGUAGES

Italian Native
English B2

HOBBIES

Gym Books
Comics Video Games

PROFESSIONAL EXPERIENCE

Software Development Specialist

Jul 2023 – Present

Engineering Ingegneria Informatica · Naples, Italy

- Own front-end implementation decisions and technology selection for Angular cybersecurity platforms developed within EU-funded research projects.
- Collaborate in English with international consortia and present implemented features, technical progress, and proposed solutions during project meetings.
- Investigated redundant calculations in an existing system and guided an optimization that improved API response times without changing the expected results.
- Organize development work through GitLab Issues and helped simplify a heavily branched Git history into a clearer commit chain, making the evolution of implemented features easier to trace.
- Coordinate the development work of up to two colleagues and mentor them through solution analysis, trade-off evaluation, and context-aware technical decisions.
- Design maintainable reactive applications with Angular, TypeScript, RxJS, NgRx, Nx, REST APIs, and WebSocket integrations.

Software Engineer

Jun 2020 – Jul 2023

NTT DATA · Naples, Italy

- Developed an Angular platform supporting administrative processes for citizens, businesses, and tax assistance centres within an Italian province.
- Modernized legacy form workflows by introducing Angular reactive forms and FormControl-based validation.
- Created reusable modules and shared components that reduced duplicated implementation work and accelerated the delivery of similar user interfaces.
- Worked directly with the client to translate complex administrative requirements into clear, easy-to-complete interfaces and coordinated a three-person development team.
- Designed the front-end integration of a major new module, defining actors, permissions, restricted application areas, and reusable versus domain-specific components.

SELECTED PROJECTS

AVANT

2025 – Present

Engineering Ingegneria Informatica

NextGenerationEU cybersecurity project for Digital Twin systems. Designed and developed workflows to monitor a Digital Twin, model business cases and connected assets, simulate MITRE ATT&CK attack paths and countermeasures, and compare inherent and residual asset risks. Researched the Digital Twin domain to translate emerging concepts into an implementable platform solution.

Technologies: Angular 22, TypeScript, RxJS, NgRx, Nx, WebSocket, Spring Boot, MongoDB, Docker

SELECTED PROJECTS

ENCRYPT

2023 – 2025

Engineering Ingegneria Informatica

Horizon Europe privacy risk assessment project developed within a consortium of organizations from across Europe. Built a web module that models domain-specific asset taxonomies, evaluates LINDDUN privacy impacts and exposure to MITRE ATT&CK techniques, calculates privacy risk scores, and summarizes assessment results through interactive charts.

Technologies: Angular 18, TypeScript, RxJS, NgRx, Nx, Tailwind CSS, Node.js, MongoDB, Docker

PAB Goes Digital

2020 – 2023

NTT DATA

Large Angular platform for digitizing provincial administrative processes used by citizens, businesses, and tax assistance centres. Introduced reactive forms, reusable modules, role-based access patterns, and shared components while working directly with the client in an Agile Scrum team.

Technologies: Angular, TypeScript, Reactive Forms, Spring Boot, MySQL

Cicciotecchio.dev

Personal project

Personal website used to experiment with modern web technologies and evolving front-end approaches. Originally developed with Angular, later rebuilt with React, and currently implemented with Next.js. Deployed on Vercel to gain hands-on experience with continuous integration and deployment workflows.

Technologies: Next.js, React, Angular, TypeScript, Vercel, CI/CD

[GitHub repository](#)

EDUCATION

Master of Science in Computer Science

2018 – 2020

Università degli Studi di Salerno · Salerno, Italy

Grade: 110/110 cum laude

Bachelor of Science in Computer Science

2013 – 2017

Università degli Studi di Salerno · Salerno, Italy

Grade: 94/110

ACADEMIC PROJECTS

Knoxly

Master's thesis privacy assistant for Telegram Web and Gmail. Built a two-stage text-classification pipeline using multilingual sentence embeddings and topic-specific sensitivity classifiers, exposed through a Flask API and integrated with a configurable Chrome extension.

Python, TensorFlow Hub, Multilingual Universal Sentence Encoder, Text Classification, Flask REST API, Chrome Extension

ShareErasmus

Erasmus platform developed while acting as project manager for a team of eight people.

Node.js, MySQL, Firebase, Jest, AWS

YASPL3

Designed a custom programming language and implemented its compiler in Java SE.

Java SE, Compilers, Language Design

N-body Simulation

Parallel and concurrent N-body simulation with AWS cluster configuration and performance testing.

C, OpenMP, AWS EC2

PUBLICATIONS

Ex Machina: Analytical Platforms, Law and the Challenges of Computational Legal Science

2018

Future Internet, MDPI · Academic research contribution

Authors: Nicola Lettieri, Antonio Altamura, Rosalba Giugno, Alfonso Guarino, Delfina Malandrino, Alfredo Pulvirenti, Francesco Vicidomini, Rocco Zaccagnino

Implementation of a Tool for Risk Assessment of Personal Data Processing Using a Methodology Based on the LINDDUN and MITRE ATT&CK Frameworks

2025

ENCRYPT Project · Technical article · Engineering Ingegneria Informatica

Author: Francesco Vicidomini